

Exam P Formula Sheet

Updated June 2026

Every formula below is derived and explained in full in the Exam P Study Manual. This page exists purely for fast lookup and final-week memorization, condensed to fit a short printout. Nothing here is a substitute for understanding where each formula comes from.

1. General Probability (23-30%)

COMPLEMENT RULE

$$P(A^c) = 1 - P(A)$$

ADDITION RULE (GENERAL)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

ADDITION RULE (MUTUALLY EXCLUSIVE)

$$P(A \cup B) = P(A) + P(B)$$

INCLUSION-EXCLUSION (3 EVENTS)

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

CONDITIONAL PROBABILITY

$$P(A | B) = P(A \cap B) / P(B), P(B) > 0$$

MULTIPLICATION RULE

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

INDEPENDENCE

$$P(A \cap B) = P(A) \cdot P(B) \Leftrightarrow P(A|B) = P(A)$$

LAW OF TOTAL PROBABILITY

$$P(A) = \sum_i P(A|B_i) \cdot P(B_i), \text{ for a partition } \{B_1, \dots, B_n\}$$

BAYES' THEOREM

$$P(B_j|A) = [P(A|B_j) \cdot P(B_j)] / \sum_i P(A|B_i) \cdot P(B_i)$$

PERMUTATIONS

$${}_nP_r = n! / (n - r)!$$

COMBINATIONS

$${}_nC_r = n! / [r!(n - r)!]$$

MULTINOMIAL COEFFICIENT

$$n! / (n_1!n_2!\dots n_k!), \sum n_i = n$$

Which Counting Tool?

Order matters?	Repetition allowed?	Tool
Yes	No	Permutations, ${}_nP_r$
Yes	Yes	n^r (multiplication principle)
No	No	Combinations, ${}_nC_r$
No	Yes	Stars and bars, $(n+r-1 \text{ choose } r)$

2. Univariate Random Variables (44-50%)

EXPECTED VALUE

$E[X] = \sum_x x \cdot f(x)$ (discrete) or $\int x \cdot f(x) dx$ (continuous)

LAW OF THE UNCONSCIOUS STATISTICIAN

$E[g(X)] = \sum_x g(x) \cdot f(x)$ or $\int g(x) \cdot f(x) dx$

VARIANCE (SHORTCUT)

$\text{Var}(X) = E[X^2] - (E[X])^2$

LINEAR TRANSFORMATION

$E[aX+b] = aE[X]+b$, $\text{Var}(aX+b) = a^2\text{Var}(X)$

COEFFICIENT OF VARIATION

$CV(X) = \sigma/\mu$, $\mu \neq 0$

LIMITED EXPECTED VALUE

$E[X \wedge u] = \int_0^u x f(x) dx + u \cdot P(X > u) = \int_0^u [1-F(x)] dx$

PAYMENT R.V. (DEDUCTIBLE D, COINSURANCE A, LIMIT U)

$Y = \alpha \cdot [\min(X, u) - \min(X, d)] = \alpha \cdot [(X \wedge u) - (X \wedge d)]$

Discrete Distributions

DISCRETE DISTRIBUTION SUMMARY

Distribution	pmf $f(x)$	$E[X]$	$\text{Var}(X)$	Notes
Bernoulli(p)	$p^x (1-p)^{1-x}$	p	$p(1-p)$	$x \in \{0, 1\}$
Binomial(n,p)	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$	sum of n iid Bernoulli(p)
Geometric(p) — trials to 1st success	$(1-p)^{x-1} p$	$1/p$	$(1-p)/p^2$	$x=1, 2, \dots$; memoryless
Geometric(p) — failures before success	$(1-p)^x p$	$(1-p)/p$	$(1-p)/p^2$	$x=0, 1, \dots$; watch parameterization!
Negative Binomial(r,p)	$\binom{x-1}{r-1} p^r (1-p)^{x-r}$	r/p	$r(1-p)/p^2$	sum of r iid Geometrics
Hypergeometric(N,K,n)	$\frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$	nK/N	$n(K/N)(1-K/N) [(N-n)/(N-1)]$	sampling w/o replacement
Poisson(λ)	$e^{-\lambda} \lambda^x / x!$	λ	λ	sum of independent Poissons adds λ 's
Discrete Uniform $\{1, \dots, n\}$	$1/n$	$(n+1)/2$	$(n^2-1)/12$	equally likely integers

Continuous Distributions

CONTINUOUS DISTRIBUTION SUMMARY

Distribution	pdf $f(x)$	$E[X]$	$\text{Var}(X)$	Notes
Uniform(a,b)	$1/(b-a)$	$(a+b)/2$	$(b-a)^2/12$	$a \leq x \leq b$
Exponential(λ)	$\lambda e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$	$x > 0$; memoryless; $\lambda = \text{rate}$

Gamma(α, λ)	$\lambda^\alpha x^{\alpha-1} e^{-\lambda x} / \Gamma(\alpha)$	α/λ	α/λ^2	sum of α iid Exp(λ) when $\alpha \in \mathbb{Z}^+$
Normal(μ, σ^2)	$(1/\sqrt{2\pi\sigma^2}) e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2	standardize $Z = (X-\mu)/\sigma$
Beta(a, b)	$x^{a-1}(1-x)^{b-1}/B(a, b)$	$a/(a+b)$	$ab/[(a+b)^2(a+b+1)]$	$0 \leq x \leq 1$; models proportions

3. Multivariate Random Variables (23-30%)

MARGINAL PMF

$$f_X(x) = \sum_y f(x, y)$$

CONDITIONAL PMF

$$f(x|y) = f(x, y) / f_Y(y)$$

INDEPENDENCE (JOINT)

$$f(x, y) = f_X(x) \cdot f_Y(y) \text{ for all } x, y$$

COVARIANCE

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

CORRELATION

$$\rho = \text{Cov}(X, Y) / (\sigma_X \sigma_Y), -1 \leq \rho \leq 1$$

VARIANCE OF A SUM

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \cdot \text{Cov}(X, Y)$$

VARIANCE OF A SUM (INDEPENDENT)

$$\text{Var}(\sum X_i) = \sum \text{Var}(X_i)$$

LAW OF TOTAL EXPECTATION

$$E[X] = E[E[X|Y]]$$

LAW OF TOTAL VARIANCE

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y])$$

MAX ORDER STATISTIC (N IID)

$$F_{(n)}(x) = [F(x)]^n, f_{(n)}(x) = n[F(x)]^{n-1}f(x)$$

MIN ORDER STATISTIC (N IID)

$$F_{(1)}(x) = 1 - [1 - F(x)]^n, f_{(1)}(x) = n[1 - F(x)]^{n-1}f(x)$$

GENERAL ORDER STATISTIC PDF

$$f_{(k)}(x) = [n! / ((k-1)!(n-k)!)] \cdot [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x)$$

CENTRAL LIMIT THEOREM

$$(\bar{X} - \mu) / (\sigma / \sqrt{n}) \rightarrow N(0, 1) \text{ as } n \rightarrow \infty$$

CONTINUITY CORRECTION

$$P(X \leq k) \approx \Phi[(k + 0.5 - \mu) / \sigma]$$

Reproductive ("Sum-Stable") Families — Independent X_i

If $X_1, \dots, X_n$ are independent...	...then the sum is
Binomial(n_i, p) — common p	Binomial($\sum n_i, p$)
Poisson(λ_i)	Poisson($\sum \lambda_i$)
Normal(μ_i, σ_i^2)	Normal($\sum \mu_i, \sum \sigma_i^2$)
Gamma(α_i, λ) — common λ	Gamma($\sum \alpha_i, \lambda$)
Exponential(λ) — n of them	Gamma(n, λ)

4. Moment Generating Functions

Not always tested directly, but useful for confirming sums of independent random variables (a repeated MGF-product argument underlies every row of the "reproductive families" table above): if X, Y independent, $M_{\{X+Y\}}(t) = M_X(t) \cdot M_Y(t)$.

BERNOULLI(P) $M(t) = (1-p) + pe^t$	BINOMIAL(N,P) $M(t) = [(1-p) + pe^t]^n$
GEOMETRIC(P) — TRIALS TO SUCCESS $M(t) = pe^t / [1-(1-p)e^t], t < -\ln(1-p)$	POISSON(λ) $M(t) = \exp[\lambda(e^t-1)]$
UNIFORM(A,B) $M(t) = (e^{tb} - e^{ta}) / [t(b-a)], t \neq 0$	EXPONENTIAL(λ) $M(t) = \lambda/(\lambda-t), t < \lambda$
GAMMA(A,λ) $M(t) = [\lambda/(\lambda-t)]^a, t < \lambda$	NORMAL(μ, σ^2) $M(t) = \exp[\mu t + \sigma^2 t^2 / 2]$

5. Standard Normal Table ($\Phi(z)$)

On the real exam, always use the official table under the Exhibit button rather than a memorized abbreviated table — this is for quick sanity-checking only.

$\Phi(Z) = P(Z \leq Z)$ FOR SELECTED Z

z	$\Phi(z)$	z	$\Phi(z)$
0.00	0.5000	1.28	0.8997
0.25	0.5987	1.40	0.9192
0.50	0.6915	1.645	0.9500
0.67	0.7486	1.75	0.9599
0.75	0.7734	1.96	0.9750
0.84	0.7995	2.00	0.9772
1.00	0.8413	2.33	0.9901
1.04	0.8508	2.576	0.9950
1.15	0.8749	3.00	0.9987

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Exam P Formula Sheet — condensed from the Exam P Study Manual's Appendix A. For derivations and worked examples, see the full study manual.